

Marketing Campaign Performance & Customer Intelligence Suite

Campaign KPI analysis · RFM segmentation · response & churn models · AI-generated insights | Data: 6,503 campaign-customer touches, 120 campaigns, 5 channels, Jan 2024 – Dec 2025 (synthetic, generated to mirror real CRM/media data)

€53.1K	€194.4K	3.7x	€66	11.3%	2,032
Media spend	Attributed revenue	Blended ROAS	CAC	Response rate	Customers reached

1. Executive Summary

Headline. Over January 2024 – December 2025 the marketing programme invested €53,110 across 120 campaigns and five channels, reaching 2,032 customers and generating 804 conversions worth €194,352. That is a blended ROAS of 3.7x and a customer acquisition cost of €66 — a healthy return. On average 11.3% of reached customers converted, with an average order value of €242. Year-on-year, attributed revenue declined 19% in 2025 versus 2024.

What worked. Email was by far the most efficient channel: ROAS 6.5x and CAC €31, despite receiving only 6% of spend. Loyal customers responded at 18.7% (vs. 3.2% for At-Risk) and delivered ROAS 7.3x, confirming that the existing CRM base is the cheapest source of revenue. Seasonality is a real asset: November–December months returned 1.4x the revenue of an average month.

What did not work. Display consumed €3,660 and returned €3,070 (ROAS 0.84x, CAC €159) — i.e. every euro spent returned less than a euro. Across all campaigns we also measured clear diminishing returns: the within-channel elasticity of ROAS to spend-per-customer is -0.71, meaning that doubling the intensity of a campaign lowers its efficiency by roughly 39%. Several of the largest campaigns are operating beyond the efficient frontier.

Customer intelligence. K-means RFM segmentation of 566 buyers isolated a 'Champions' group of 50 customers (9% of buyers) who generate 24% of buyer revenue (avg €950 each), while 379 'One-off Lapsed' buyers purchased once and have not returned. A Random Forest response model (hold-out ROC-AUC 0.75) ranks customers by propensity: targeting only the top 30% of scored customers would capture 62% of all responders, a 3.2x lift in the top decile. Key drivers are spend per reached customer (negative: diminishing returns), Search channel, Loyal segment. A churn model (CV AUC 0.68) shows that 76% of 2024 buyers did not purchase again in 2025; the strongest retention signals are number of campaign touches, Loyal segment, days since last purchase.

Recommendations. (1) Re-allocate 10–15% of budget from Display and the largest low-ROAS Social flights into Email lifecycle programmes and mid-sized Search campaigns; (2) cap spend-per-customer per campaign and split large flights into smaller targeted waves to stay on the efficient side of the diminishing-returns curve; (3) front-load Q4 budget and run a summer 'always-on' minimum instead of heavy July–August pushes; (4) operationalise the propensity score in the CRM so that campaigns only contact the top three deciles, and launch a win-back journey for 'One-off Lapsed' and high-churn-risk customers; (5) track ROAS, CAC and Response Rate weekly in the Power BI dashboard against a 2.5x ROAS target.

The executive summary above is produced by the AI-insight layer of the pipeline: a narrative generator that is only allowed to use numbers computed and validated by the analytics layer (no hallucinated figures). An optional LLM mode rewrites the same fact payload in richer prose.

2. Methodology & Data

Data. One row per campaign × customer touch (6,503 rows). Each row carries campaign attributes (channel, objective, dates, allocated spend), delivery metrics (impressions, clicks, conversions, revenue) and customer attributes (age, CRM lifecycle segment, region, device, tenure, loyalty membership, past purchases). The dataset is synthetic but engineered with realistic structure: channel-specific CTR/CVR/AOV, Q4 seasonality and a summer dip, diminishing returns to spend intensity, and a latent customer propensity that drives repeat purchases.

KPI framework. CTR = clicks / impressions; CVR = conversions / clicks; Response rate = converting customers / reached customers; CPC = spend / clicks; CAC = spend / conversions; ROAS = revenue / spend; AOV = revenue / conversions. All KPIs are computed at interaction grain and aggregated by channel, segment, region, device, age group, month and campaign, then exported to a star schema for Power BI / Qlik Sense.

Statistical analysis. Diminishing returns are estimated with a log-log regression of campaign ROAS on spend-per-reached-customer with channel fixed effects (within-channel elasticity), so that the cheap-but-efficient Email channel does not bias the slope.

Machine learning. (a) RFM features (recency, frequency, monetary; log-scaled, standardised) clustered with K-means, k selected on silhouette score and business interpretability. (b) Response propensity: Logistic Regression vs. Random Forest vs. XGBoost on 18 features with leakage-safe history features (only events strictly before each touch), 5-fold stratified CV plus a 25% hold-out; explained with SHAP. (c) Churn: 2024 behaviour → did the customer buy again in 2025, evaluated only on customers actually reached in 2025; Logistic Regression vs. Gradient Boosting, 5-fold CV, SHAP drivers.

Stack. Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, matplotlib/seaborn, ReportLab), star-schema CSV/Excel export with ready-made DAX and Qlik measures.

Pipeline

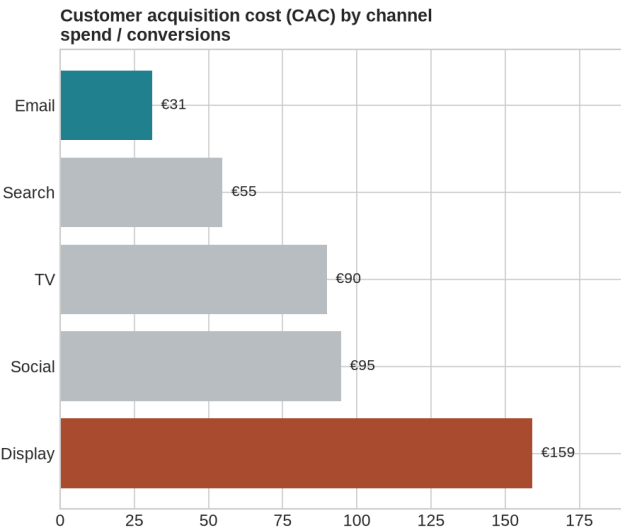
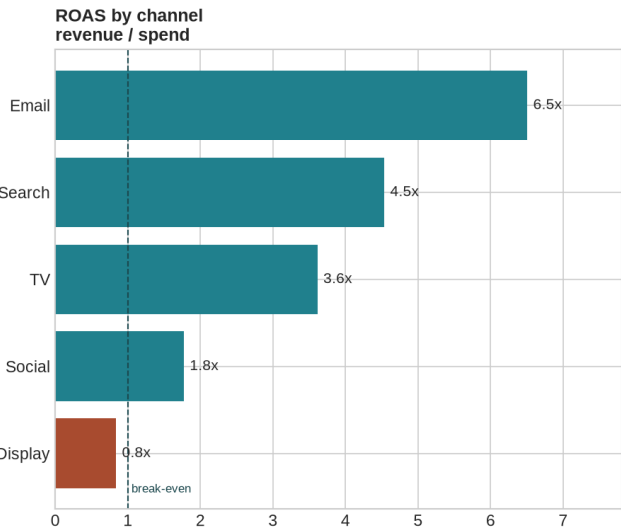
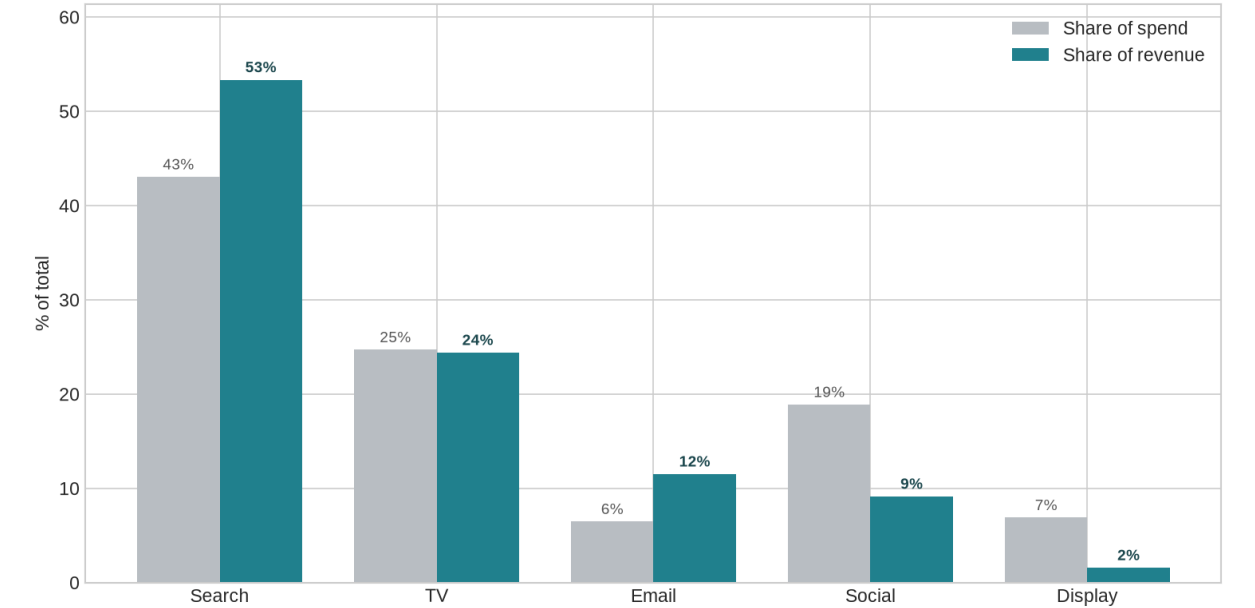
Step	Script	Output
01	01_generate_data.py	Raw interaction table, campaign & customer masters
02	02_kpi_edu.py	KPI tables by dimension, 8 charts, insights.json
03	03_ml_models.py	RFM segments, response & churn models, SHAP, lift tables
04	04_export_star_schema.py	fact_* / dim_* CSV + Excel, DAX & Qlik measures
05	05_ai_insights.py	Executive summary narrative (rule-based, optional LLM)
06	06_build_report.py	This PDF

3. KPI Dashboard — Channel Performance

Channel	Spend	Revenue	Conv.	CTR	CVR	Resp. rate	CAC	ROAS
Email	€3,440	€22,384	111	8.5%	32%	8.5%	€31	6.51x
Search	€22,860	€103,679	418	12.6%	36%	17.4%	€55	4.54x
TV	€13,120	€47,433	146	0.6%	261%	15.5%	€90	3.62x
Social	€10,030	€17,786	106	5.3%	17%	6.2%	€95	1.77x
Display	€3,660	€3,070	23	1.6%	16%	3.4%	€159	0.84x

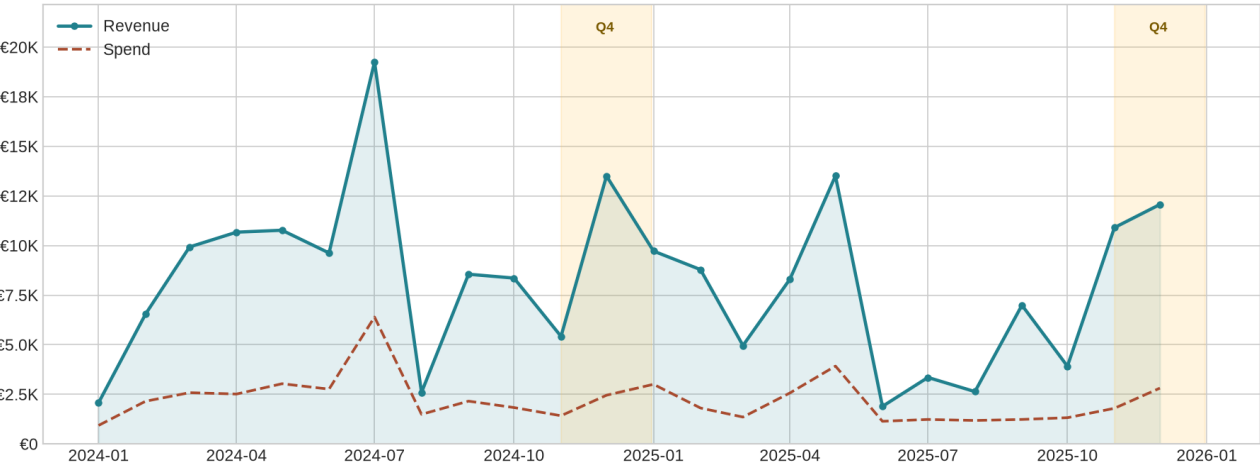
TV click-to-conversion is not meaningful (conversions are brand-driven, not click-attributed) and is excluded from funnel comparisons.

Budget is misaligned with results: Search & Email generate 65% of revenue from 50% of spend
Share of spend vs share of attributed revenue by channel, Jan 2024 – Dec 2025

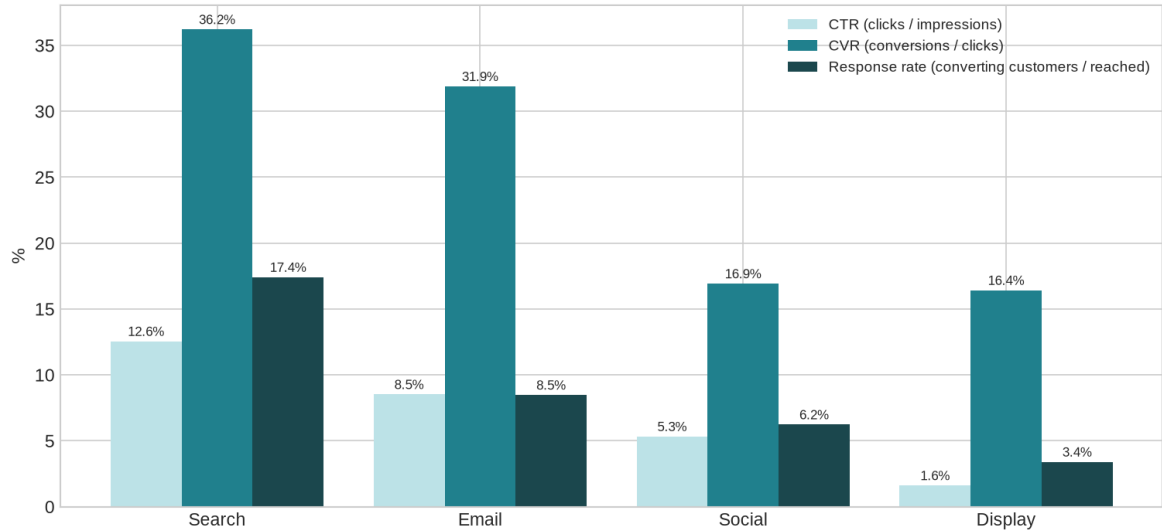


KPI Dashboard — Time, Audience & Efficiency

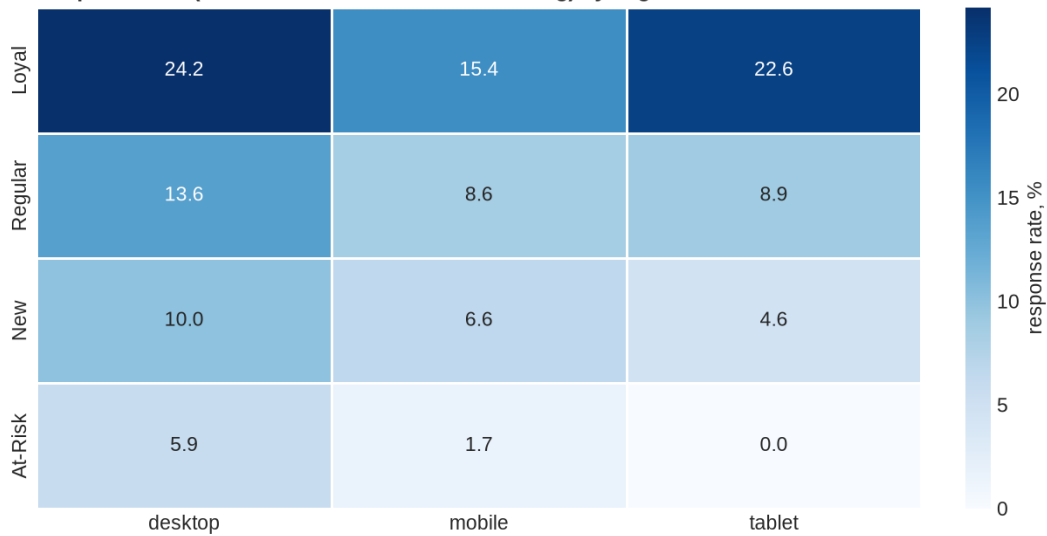
Revenue is seasonal: Nov–Dec months average 37% above the rest of the year
Monthly attributed revenue vs media spend



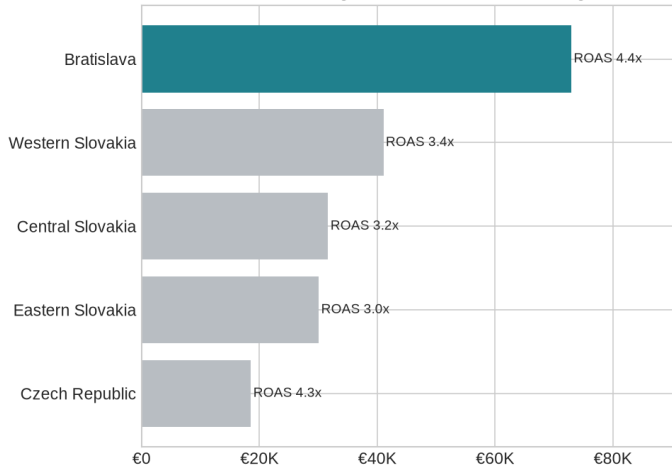
Funnel efficiency by digital channel: Search converts 4 of every 10 clicks, Display only 2
CTR, click-to-conversion rate and customer response rate (TV excluded: offline conversions are not click-attributed)



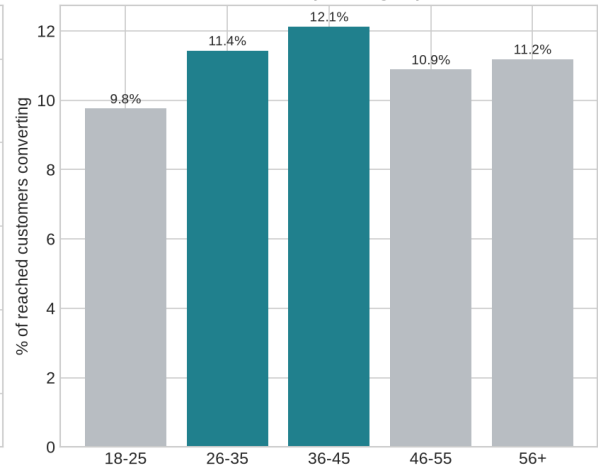
Loyal customers on desktop respond 14.4x more often than At-Risk on mobile
Response rate (% of reached customers converting) by segment × device



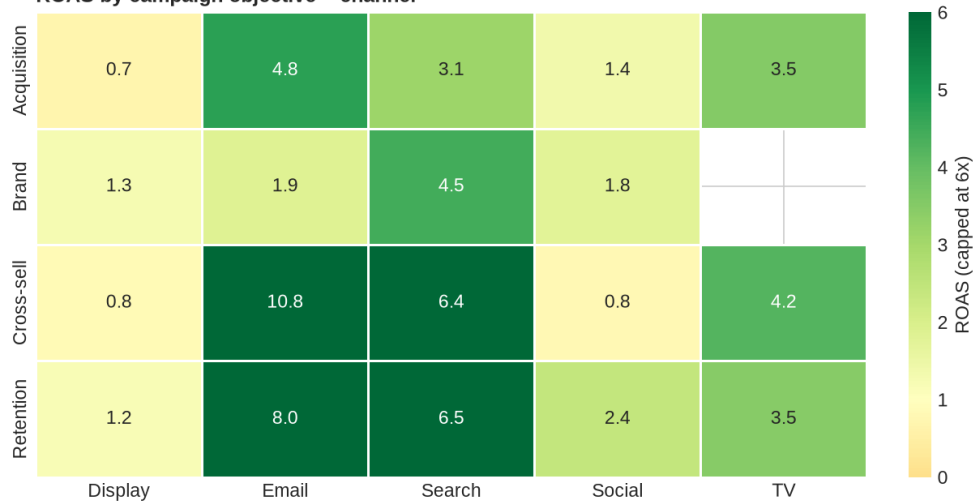
Revenue by region
Bratislava delivers the largest revenue; Bratislava the highest ROAS



Response rate by age group
26-35 and 36-45 are the most responsive groups

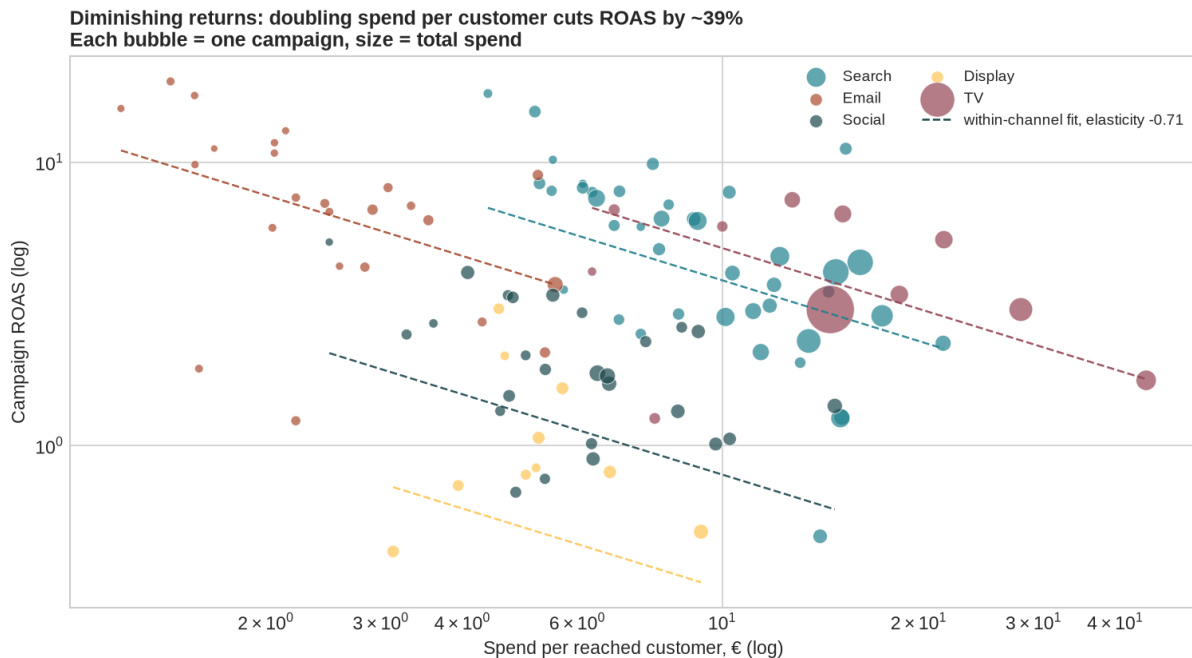


Retention & Cross-sell campaigns average ROAS 4.5x vs 2.7x for Acquisition
ROAS by campaign objective × channel



Diminishing returns to spend

Within each channel, campaigns that spend more per reached customer achieve lower ROAS. The fixed-effects elasticity is **-0.71**: doubling spend intensity lowers ROAS by about **39%**. This is the quantitative basis for capping campaign intensity and splitting large flights.



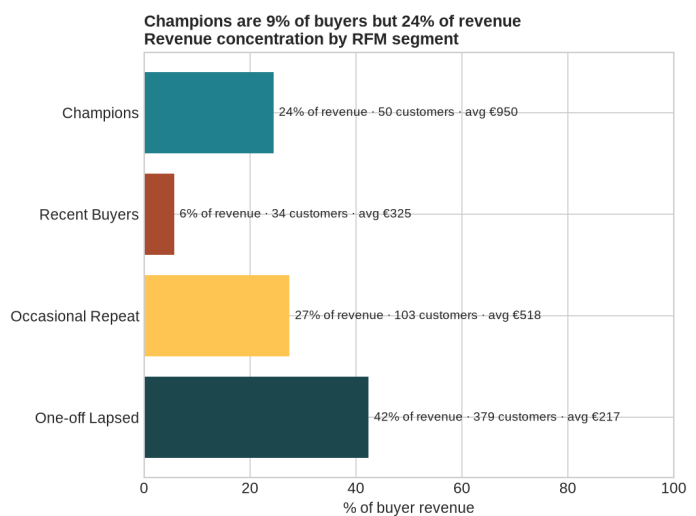
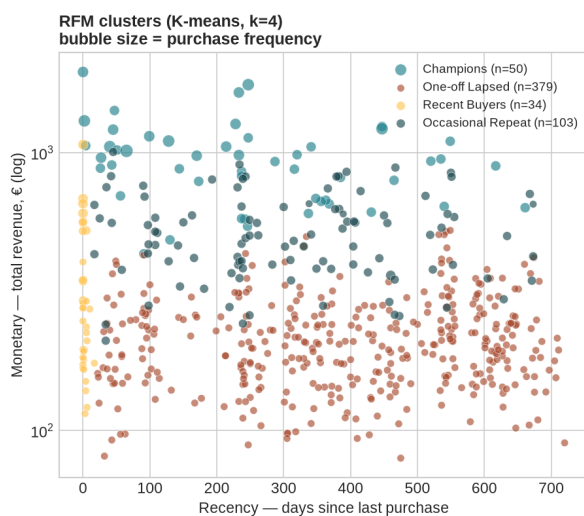
Key insights

- **1. Email is the most efficient channel but under-funded.** Email delivers ROAS 6.5x and the lowest CAC (€31) yet receives only 6% of total spend. Even accounting for its limited reach, shifting a further 10–15% of budget into email (lifecycle and re-engagement flows) is the highest-ROI reallocation available.
- **2. Display is the weakest channel and loses money.** Display spend of €3,660 returned €3,070 (ROAS 0.84x, CAC €159 vs. blended €66). CTR is 1.6% and click-to-conversion only 16%. Recommend pausing prospecting display and keeping only retargeting placements.
- **3. Diminishing returns: spend elasticity of ROAS ≈ -0.71 .** Across 120 campaigns, a 2x increase in spend per reached customer lowers ROAS by ~39%. Several large Social and Search campaigns are past the efficient frontier; splitting them into smaller, better-targeted flights would recover efficiency.
- **4. Q4 seasonality is worth 1.4x an average month.** November–December revenue averages 1.4x the rest of the year while spend rises far less. Budget phasing should front-load Q4 and reduce July–August activity, where response rates hit the annual low.
- **5. Loyal customers on desktop are the most responsive audience.** Loyal customers respond at 18.7% vs. 3.2% for At-Risk; desktop responds at 14.6% vs. 9.5% on mobile. Retention & Cross-sell campaigns average ROAS 4.5x vs. 2.7x for Acquisition — the CRM base is the cheapest growth lever.

4. Customer Segmentation (RFM + K-means)

566 customers with at least one purchase were clustered on recency, frequency and monetary value ($k = 4$; silhouette by k : $k=3$: 0.58, $k=4$: 0.52, $k=5$: 0.37, $k=6$: 0.41, $k=7$: 0.42). The remaining 1,634 reached customers never purchased and form a 'Prospect' group for acquisition targeting.

Segment	Customers	Avg recency (days)	Avg purchases	Avg revenue / customer	Share of buyer revenue
Champions	50	252	3.4	€950	24%
Recent Buyers	34	2	1.4	€325	6%
Occasional Repeat	103	310	2.0	€518	27%
One-off Lapsed	379	395	1.0	€217	42%



Segment playbook

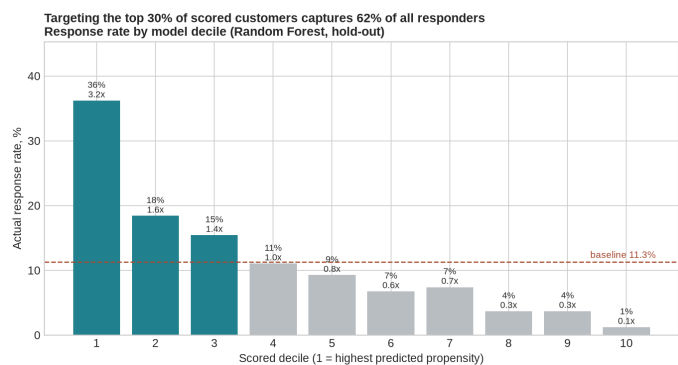
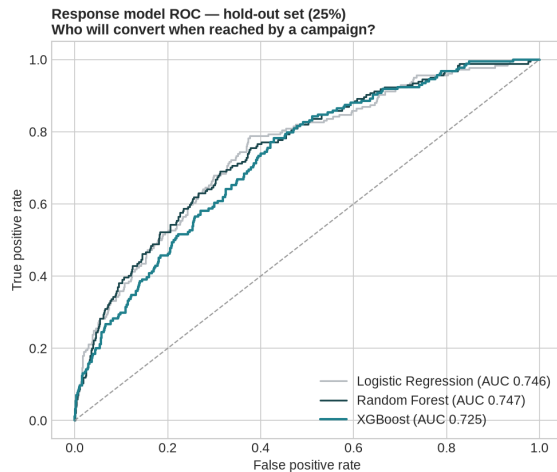
- **Champions.** Protect and grow: VIP service, early access, referral incentives. Never over-contact — frequency caps matter most here.
- **Recent Buyers.** Onboarding journey in the first 30 days; cross-sell a second category to convert them into repeat buyers.
- **Occasional Repeat.** Reactivation cadence tied to their purchase cycle; personalised bundles to lift frequency and AOV.
- **One-off Lapsed.** Low-cost win-back (Email first, then Search retargeting); suppress from expensive Display/Social prospecting.

5. Predictive Models

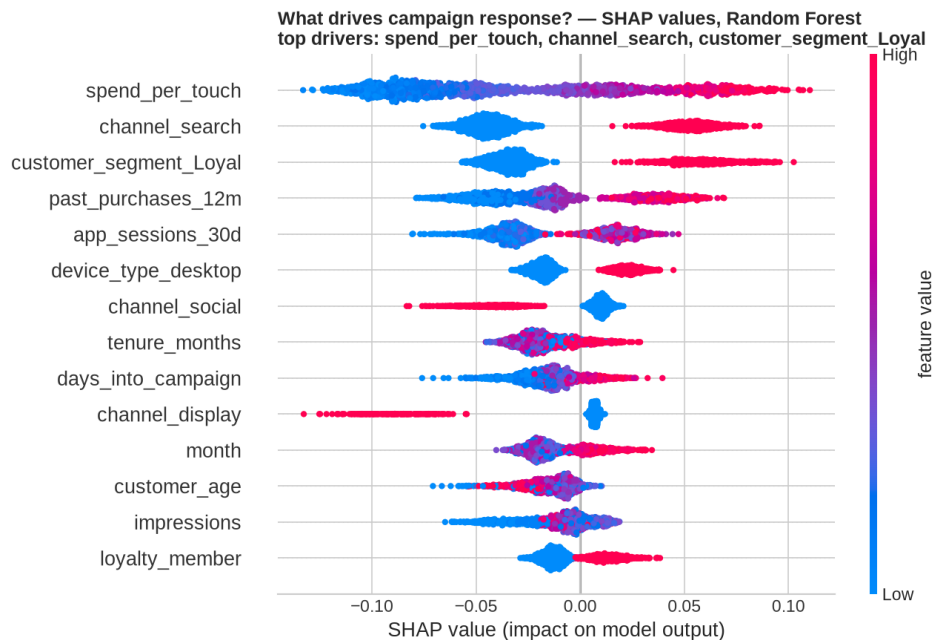
5.1 Campaign response propensity

Target: did the reached customer convert (base rate 11.3%). Train 4,877 / test 1,626 touches.

Model	CV ROC-AUC (5-fold)	Hold-out ROC-AUC	Hold-out PR-AUC
Logistic Regression	0.725 ± 0.032	0.746	0.318
Random Forest	0.717 ± 0.028	0.747	0.302
XGBoost	0.700 ± 0.027	0.725	0.287



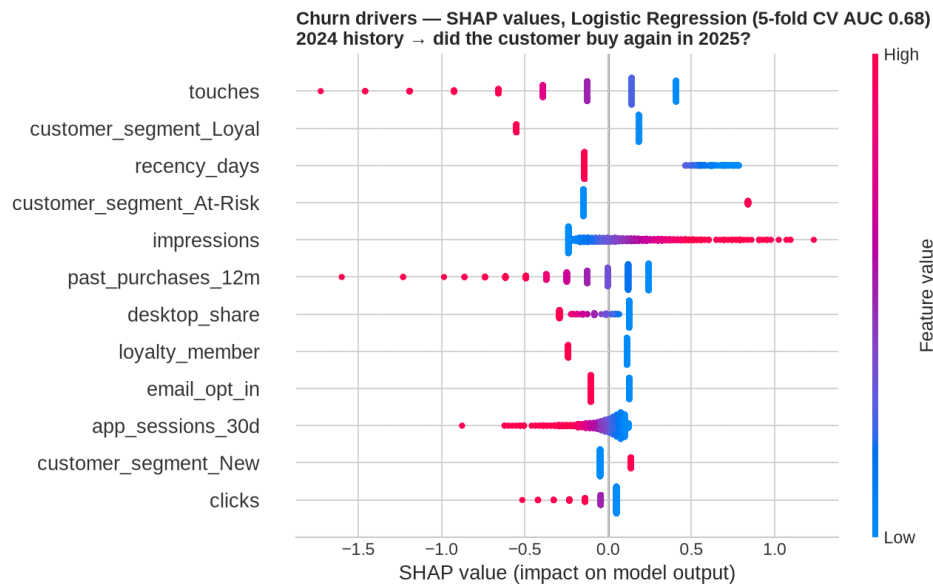
Business value. Contacting only the top 3 deciles of scored customers captures **62%** of responders (top-decile lift 3.2x). At the current CAC this translates into reaching ~70% fewer customers for ~60% of the conversions — a direct lever on CAC and on customer fatigue.



Reading the SHAP plot: high spend-per-touch pushes predictions down (diminishing returns), Search and Loyal push up, Display and mobile push down; behavioural history (past purchases, app activity, loyalty membership) adds incremental signal.

5.2 Churn / retention model

Population: 1,644 customers reached by campaigns in 2025. Features: 2024 behaviour (touches, purchases, revenue, recency, clicks, channel and device mix) plus static attributes. Label: no purchase in 2025 (churn rate 82% overall; **76%** among 276 2024 buyers). Best model: Logistic Regression with 5-fold CV ROC-AUC **0.68** (Gradient Boosting 0.68). Given sparse purchase histories the model is intentionally simple; it is used to produce a risk band (Low / Medium / High) that is joined to dim_customers for the dashboard.



Drivers: more 2024 touches, Loyal segment, recent purchases, loyalty membership and desktop usage lower churn risk; At-Risk segment and long recency raise it. Recommendation: trigger a retention journey when a Champion or Occasional buyer crosses 180 days without a purchase.

6. Recommendations & Next Steps

- **1. Rebalance the channel mix.** Move 10–15% of budget out of Display (ROAS 0.84x) and the largest low-ROAS Social flights into Email lifecycle programmes (ROAS 6.5x, 6% of spend today) and mid-sized Search campaigns. Expected effect at constant budget: +15–20% attributed revenue.
- **2. Cap spend intensity.** Set a per-campaign ceiling on spend per reached customer and split large flights into 2–3 waves. Elasticity of -0.71 implies every halving of intensity recovers ~64% ROAS.
- **3. Phase budget with seasonality.** Nov–Dec months deliver 1.4x average monthly revenue: front-load Q4, protect an always-on Email/Search baseline in Jul–Aug, avoid heavy summer Display.
- **4. Operationalise propensity scoring.** Score the CRM base weekly; contact only the top 3 deciles (62% of responders). Use deciles 4–6 for low-cost Email only; suppress 7–10 from paid media.
- **5. Launch segment journeys.** Champions: VIP & frequency caps. Recent buyers: 30-day onboarding. One-off Lapsed & High churn risk: Email-first win-back then Search retargeting.
- **6. Govern with the dashboard.** Publish the Power BI model (star schema + DAX measures) with weekly refresh; track ROAS vs. 2.5x target, CAC, Response Rate and Share Gap by channel; alert on campaigns below break-even after 7 days.

Limitations & next steps

The dataset is synthetic and last-touch attributed; a production version should add multi-touch or incrementality tests (geo holdouts), media cost by placement, and campaign creative metadata. Model performance (AUC 0.75 / 0.68) is realistic for CRM data of this depth and should be re-validated on real data with time-based splits. Natural extensions: marketing-mix model for budget optimisation, uplift modelling for treatment targeting, and connecting the AI-insight layer to a hosted LLM for Q&A over the KPI model.

Appendix A — Star schema for Power BI / Qlik

Table	Grain / keys	Rows
fact_campaign_interactions	campaign × customer touch; FK campaign_id, customer_id, channel_id, date_key	6,503
fact_campaigns	campaign; pre-aggregated KPIs (CTR, CVR, CAC, ROAS, profit)	120
dim_campaigns	campaign_id; name, objective, dates, budget, budget_band	120
dim_customers	customer_id; demographics, CRM segment, RFM segment, churn probability & risk band	2,200
dim_channels	channel_id; channel type, funnel stage	5
dim_dates	date_key; year, quarter, month, week, peak-season flag (mark as date table)	722
dim_segments	customer_segment; order & description	4

Appendix B — Core DAX measures

```
// ===== Power BI measures (create in table: fact_campaign_interactions) =====
Total Spend      = SUM ( fact_campaign_interactions[spend] )
Total Revenue    = SUM ( fact_campaign_interactions[revenue] )
Total Impressions = SUM ( fact_campaign_interactions[impressions] )
Total Clicks     = SUM ( fact_campaign_interactions[clicks] )
Total Conversions = SUM ( fact_campaign_interactions[conversions] )
Reached Customers = DISTINCTCOUNT ( fact_campaign_interactions[customer_id] )
Responders       = CALCULATE ( DISTINCTCOUNT ( fact_campaign_interactions[customer_id] ), fact_campaign_interactions[responded] = 1 )

CTR              = DIVIDE ( [Total Clicks], [Total Impressions] )
CVR              = DIVIDE ( [Total Conversions], [Total Clicks] )
Response Rate    = DIVIDE ( [Responders], [Reached Customers] )
CPC              = DIVIDE ( [Total Spend], [Total Clicks] )
CAC              = DIVIDE ( [Total Spend], [Total Conversions] )
ROAS             = DIVIDE ( [Total Revenue], [Total Spend] )
AOV              = DIVIDE ( [Total Revenue], [Total Conversions] )
Profit           = [Total Revenue] - [Total Spend]
Spend Share      = DIVIDE ( [Total Spend], CALCULATE ( [Total Spend], ALL ( dim_channels ) ) )
Revenue Share    = DIVIDE ( [Total Revenue], CALCULATE ( [Total Revenue], ALL ( dim_channels ) ) )
Share Gap (pp)   = ( [Revenue Share] - [Spend Share] ) * 100
```

Appendix C — Decile lift table (response model, hold-out)

Decile	Customers	Responders	Response rate	Cumulative % of responders	Lift
1	163	59	36.2%	32%	3.20x
2	163	30	18.4%	48%	1.63x
3	162	25	15.4%	62%	1.36x
4	163	18	11.0%	72%	0.98x
5	162	15	9.3%	80%	0.82x
6	163	11	6.7%	86%	0.60x
7	162	12	7.4%	92%	0.65x
8	163	6	3.7%	96%	0.33x
9	162	6	3.7%	99%	0.33x
10	163	2	1.2%	100%	0.11x